

# HEEJUN KIM

 [Homepage](#)  [Email](#)  [GitHub](#)  [Google Scholar](#)

## EDUCATION

### Gwangju Institute of Science and Technology (GIST)

February 2020 - present

B.S Major: Electrical Engineering and Computer Science, GPA 4.21/4.50

Location: Gwangju, South Korea

### University of California, Berkeley (UC Berkeley)

June 2024 - August 2024

Summer Session, GPA 4.0/4.0

Coursework: Introduction to Artificial Intelligence, Principles & Techniques of Data Science

## RESEARCH EXPERIMENTS

### Learning and Adaptation Algorithm Lab

January 2026 - present

Research Assistant, Seoul National University

Seoul, South Korea

- Supervised by [Prof. Taesup Kim](#)
- Conducting research on Agentic Memory Using LLM Uncertainty Estimation

### Data Science Lab

Feb 2025 - present, January 2024 - June 2024

Research Assistant, Gwangju Institute of Science and Technology

Gwangju, South Korea

- Supervised by [Prof. Sundong Kim](#)
- Developed an architecture applying generative models to reinforcement learning for solving the Abstract Reasoning Corpus (ARC).
- Researched test-time adaptation with reinforcement learning for self-improving reasoning capabilities of large language model

### Forschungszentrum Jülich

September 2024 - February 2025

Korea-Germany Junior Research Fellowship Program 2024

Jülich, Germany

- Host Institute: INM-7 directed by [Prof. Simon B. Eickhoff](#) and contributed to Mathematical Neuroscience Group led by [Dr. Oleksandr Popovych](#).
- Developed preprocessing pipeline for multi-modal MRI human connectome data using HPC and Hyper-Threading (Project Supervisor: [Dr. Kyesam Jung](#))
- Fund: 12,000,000KRW (8,197EUR) by National Research Foundation of Korea (NRF)(No. RS-2023-00254343).

## PUBLICATIONS

### From Noise to Diversity: Random Embedding Injection in LLM Reasoning

[Heejun Kim](#), Seungpil Lee, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, Sundong Kim

[Under Review] *ICML 2026 Mechanistic Interpretability Workshop (Spotlight, 2.8% acceptance rate)*. [\[arXiv\]](#)

### Hallucination as Commitment Failure: Larger LLMs Misfire Despite Knowing the Answer

Jewon Yeom, Jaewon Sok, [Heejun Kim](#), Seonghyeon Park, Jeongjae Park, Taesup Kim

[Under Review] *ICML 2026 Mechanistic Interpretability Workshop (Poster)*. [\[arXiv\]](#)

### On the Role of Proposal Support in Diffusion-Based Offline RL for Sequential Decision-Making

Yunho Kim, Jaehyun Park, [Heejun Kim](#), Sejin Kim, Byung-Jun Lee, Sundong Kim

[Under Review] *ICML 2026 DEMO Workshop (Poster)*.

### Disentangling Question, Response, and Verifier Languages in Multilingual LLM Evaluation

Jewon Yeom, Jaewon Sok, Seonghyeon Park, [Heejun Kim](#), Jeongjae Park, Taesup Kim

[Under Review]

### On the Tip of the Tongue: First-Token Cueing as a Behavioral Probe of Recall in LLMs

Jaewon Sok, Jewon Yeom, [Heejun Kim](#), Seonghyeon Park, Jeongjae Park, Taesup Kim

[Under Review]

## Efficient Epistemic Uncertainty Estimation for Large Language Models via Knowledge Distillation

Seonghyeon Park, Jewon Yeom, Jaewon Sok, Jeongjae Park, Heejun Kim, Taesup Kim  
[Under Review] [\[arXiv\]](#)

## Diffusion-Based Offline RL for Improved Decision-Making in Augmented ARC Task

Yunho Kim, Jaehyun Park, Heejun Kim, Sejin Kim, Byung-Jun Lee, Sundong Kim  
[arXiv:2410.11324](#)

## PROJECTS

---

### Skill-based Reinforcement Learning with UCB Skill Selection

*May 2025 - June 2025*

- Reproduced BeCL framework for unsupervised skill discovery in reinforcement learning environments
- Developed Multi-Armed Bandit approach with UCB algorithm to improve downstream task adaptation efficiency. [\[Poster\]](#)

### Hyundai AI Challenge

*September 2023 - October 2023*

- Processed large-scale port logistics datasets (cleaning, feature extraction, temporal aggregation).
- Built tree-based boosting models to predict ship waiting times.
- Ranked Top 25/330 teams among 1,300+ participants.

## LEADERSHIPS

---

### Teaching Assistant

*Fall 2025*

- Machine Learning & Deep Learning (AI5213/EC4213): led TA sessions, held office hours, managed assignments/exams, and organized Kaggle competition. [\[Homepage\]](#)

### Korea Army Instructor

*February 2021 - August 2022*

- Served as an Army instructor in the 50th Division in Daegu, South Korea
- Conducted basic training for approximately 2000 military personnel

## EXTRA ACTIVITIES

---

### Google Developer Student Club

*September 2023 - February 2024*

- Participated in an AI study focused on machine learning and deep learning
- Studied models in ML/DL and analysis of top team models in Kaggle AI competitions
- Established study directions and organized study materials as a team leader

### Financial Investment Conference

*March 2023 - March 2024*

- Conducted a study on economics
- Analyzed and presented economic markets on a weekly basis

### GIST Club Federation

*February 2020 - December 2020*

- Managed clubs in Gwangju Institute of Science and Technology

## SKILLS

---

### Programming Languages

Python, C/C++

### Python Packages & Framework

PyTorch, VeRL, Scikit-learn, Pandas, Numpy, Matplotlib, Ray

### Operating System

Linux, Windows, MacOS

### Others

Docker, Wandb, SLURM, HTCondor, SQLite

## HONORS

---

|                                                                            |                                       |
|----------------------------------------------------------------------------|---------------------------------------|
| <b>Government Scholarship, Gwangju Institute of Science and Technology</b> | <i>2020 - present</i>                 |
| <b>Government Scholarship, STEM scholarship</b>                            | <i>2020 - present</i>                 |
| <b>Dean's List, College of Information and Computing, GIST</b>             | <i>Spring 2025, Fall 2025</i>         |
| <b>GIST International Internship Support Program</b>                       | <i>September 2024 - February 2025</i> |
| <b>Korea-Germany Junior Research Fellowship Program 2024</b>               | <i>September 2024 - February 2025</i> |
| · Organized by the Max Planck POSTECH/KOREA Research (MPK)                 |                                       |